Step 6 Covariate-Adjusted Trial-Level ROI Report

Overview

Metric	Value
Included participant-sessions	36
Included participants	36
Included questions	30
Included trials	1075
Primary model type	MixedLM
Primary model fallback used	no
Primary condition p-value	0.052102

Primary question. Does the abstract-versus-concrete effect in the onset-locked left frontal ROI remain after adjusting for question length and participant/question variability?

Warnings and QC. MODEL_PRIMARY_FRONT_WORDCOUNT: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_SENTENCECOUNT: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_SENTENCELENGTH: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_WORDCOUNT_DI Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_BACK_WORDCOUNT: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_WORDCOUNT_RT: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_WORDCOUNT_PARTICIPANT_CORRECT: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.

Item-Level Balance Check

Covariate	Abstract mean (SD)	Concrete mean (SD)	SMD	Welch p
Total word count	83.60 (24.85)	77.47 (42.26)	0.177	0.632644
Sentence count	3.73 (2.46)	3.67 (2.55)	0.027	0.942510
Average sentence length	29.67 (18.33)	27.33 (14.57)	0.142	0.701263
ENEM item correctness	0.43 (0.12)	0.33 (0.13)	0.770	0.044204

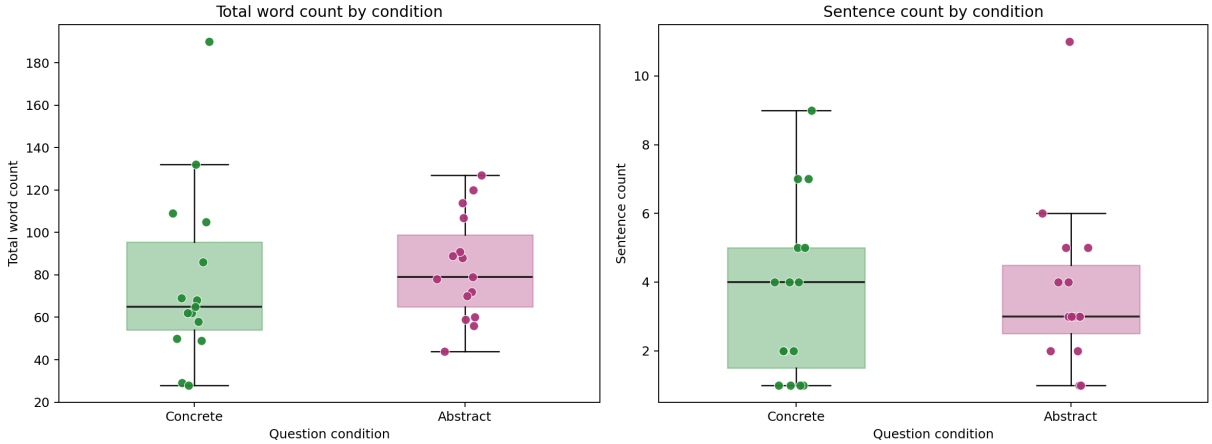

Figure 1: Item-level balance plots for total word count and sentence count by question condition.

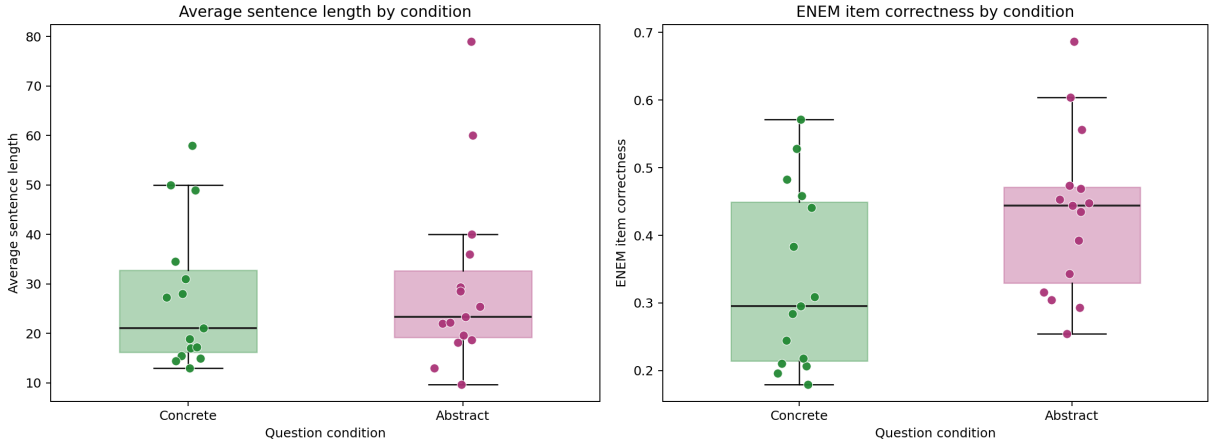

Figure 2: Item-level balance plots for average sentence length and ENEM item correctness by question condition.

Primary Front ROI Model

Metric	Value
Model type	MixedLM
Included participants	36
Included questions	30
Included trials	1075
Condition beta	3.600e-08
Condition SE	1.854e-08
Condition statistic	1.942
Condition p-value	0.052102
Condition 95% CI	[-3.276e-10, 7.233e-08]
Word-count beta	-1.200e-08
Residual variance	2.996e-14
Participant random-intercept variance	2.912e-15
Question random-intercept variance	5.903e-14

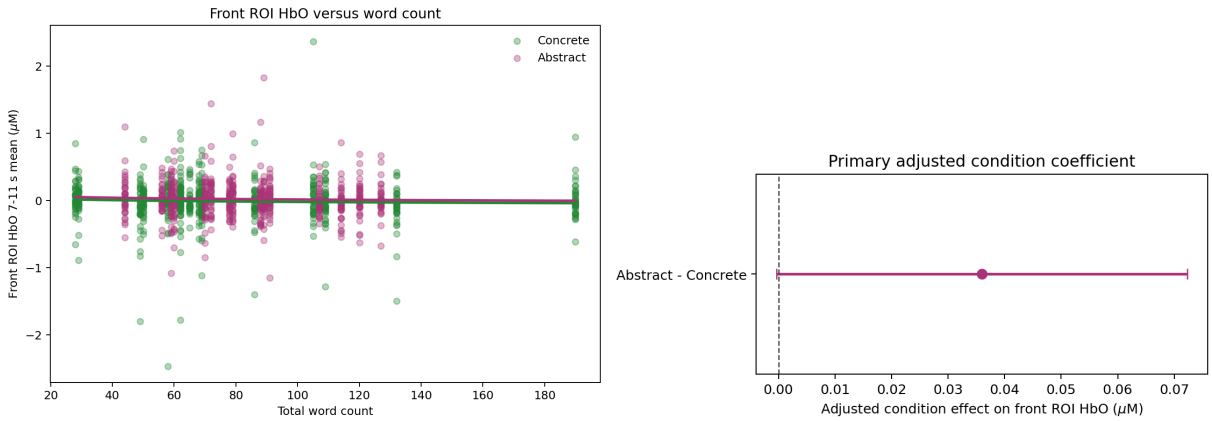

Figure 3: Front ROI HbO versus word count and the adjusted condition effect from the primary word-count model.

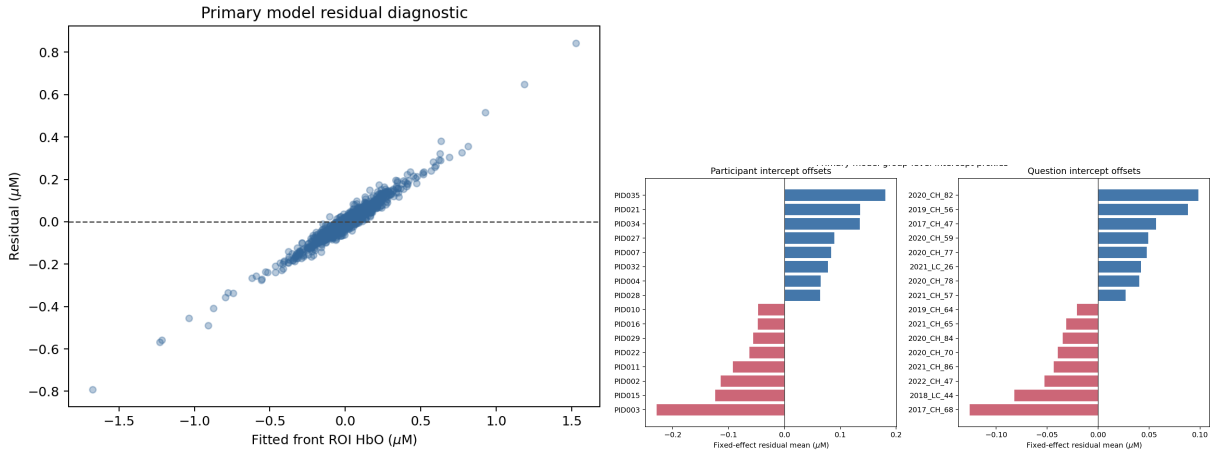

Figure 4: Residual diagnostic and group-level intercept proxies for the primary model.

Secondary and Sensitivity Models

Model	Cond. beta	Cond. p	Part. var	Question var	Fallback
Alternative front ROI adjusted for sentence count	3.159e-08	0.082763	2.908e-15	5.911e-14	no
Alternative front ROI adjusted for sentence length	3.195e-08	0.080115	2.908e-15	5.912e-14	no
Sensitivity front ROI adjusted for log word count and ENEM correctness	2.920e-08	0.145237	2.914e-15	5.898e-14	no
Secondary back ROI adjusted for log word count	1.011e-08	0.526536	3.900e-15	3.880e-14	no
Sensitivity front ROI adjusted for log word count and log response time	3.592e-08	0.052593	3.032e-15	5.895e-14	no
Sensitivity front ROI adjusted for log word count and participant correctness	3.359e-08	0.072130	2.930e-15	5.896e-14	no

Final Conclusion

In the trial-level mixed-effects analysis adjusted for log word count, the abstract-versus-concrete effect in the left frontal ROI was not statistically significant (beta=3.600e-08, SE=1.854e-08, p=0.052102, 95